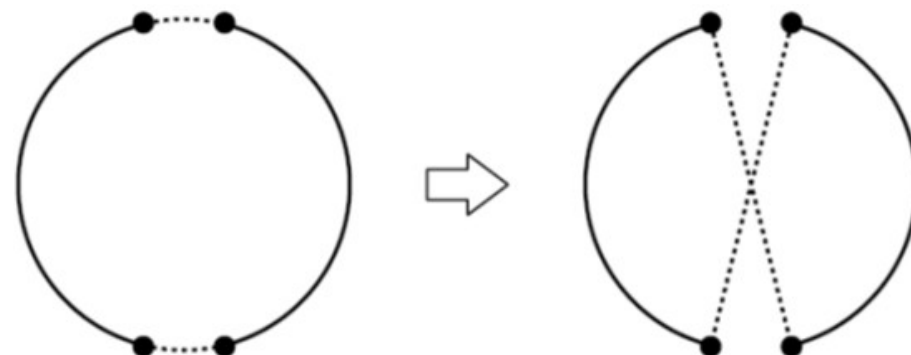
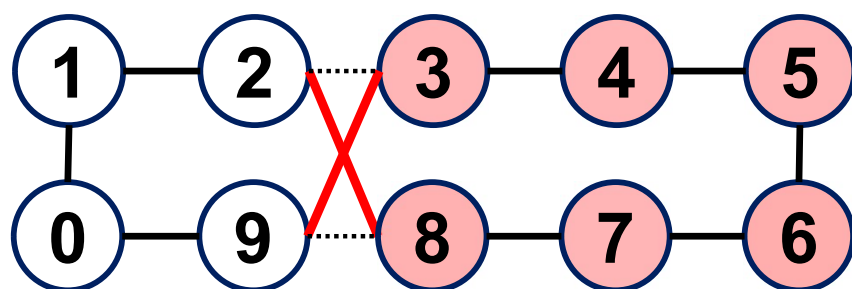


Lab Session Task 1

Local search

- Neighborhood structure: Inversion (two-edge change)



Creation of a new initial solution:

- 1st initial solution: Greedy solution

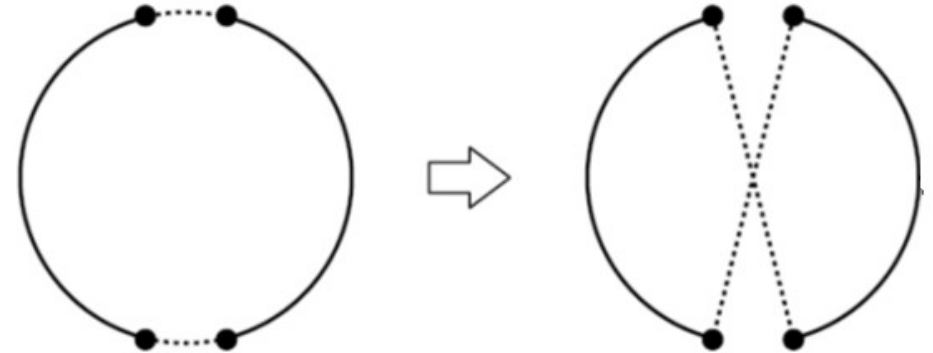
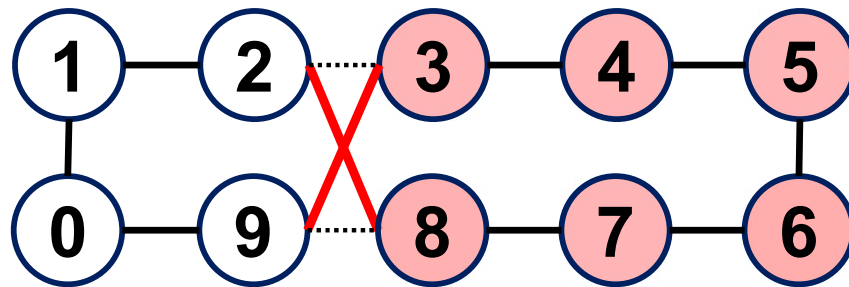
Task 1

Create a TSP problem where a **non-optimal** greedy solution is obtained from an initial city and the obtained greedy solution cannot be improved by inversion-based local search. It is enough to show that one greedy solution cannot be improved. You do not have to show that any greedy solution from an arbitrary initial city cannot be improved.

Lab Session Task 2

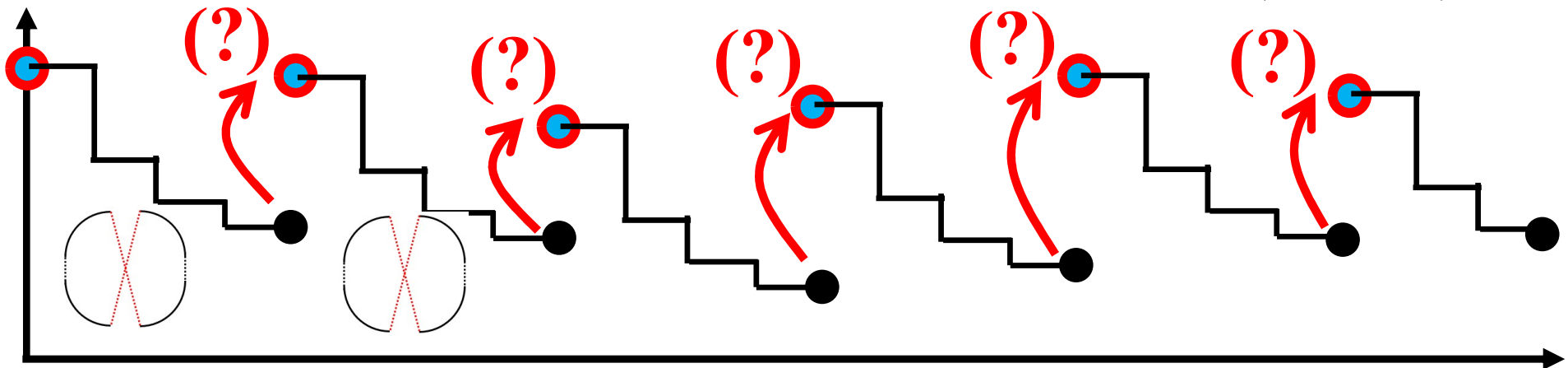
Iterated Local search

- Neighborhood structure: Inversion (two-edge change)



Creation of a new initial solution:

- 1st initial solution: Greedy solution
- Other initial solutions: Explain your own idea about how to create an initial solution for each local search run (Task 2).



Lab Session Task 3

Specify neighborhood structures $N_1, N_2, N_3, \dots$ for designing a **Variable Neighborhood Descent (VND)** algorithm for a knapsack problem.